Claudio **Roncoli**

Associate Professor · KU Leuven

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Visiting Professor · Aalto University

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https://claudioroncoli.github.io | Claudio Roncoli



Research interests

- Modelling and control of traffic systems in presence of automated and connected vehicles
- Integration, management, and optimisation of emerging mobility systems and technologies
- Optimisation of Intelligent Transportation Systems for smart and sustainable mobility
- Traffic flow theory, modelling, and simulation

Professional positions

Associate Professor — *11/2024 – current*
Mobility and Industrial Management (MIM), Faculty of Engineering Science, KU Leuven — *Leuven, Belgium*

Visiting Professor — *10/2024 – current*
Department of Built Environment, School of Engineering, Aalto University — *Espoo, Finland*

Associate Professor — *12/2022 – 10/2024*
Department of Built Environment, School of Engineering, Aalto University — *Espoo, Finland*

Visiting Researcher — *10/2023 – 12/2023*
School of Civil Engineering, University of Sydney — *Sydney, Australia*

Visiting Scholar — *09/2023 – 10/2023*
School of Civil Engineering, University of Queensland — *Brisbane, Australia*

Assistant Professor — *10/2016 – 11/2022*
Department of Built Environment, School of Engineering, Aalto University — *Espoo, Finland*

Postdoctoral Research Associate — *05/2013 – 09/2016*
Dynamic Systems & Simulation Laboratory, Technical University of Crete — *Chania, Greece*

Visiting Student — *07/2012*
IMT School for Advanced Studies — *Lucca, Italy*

Visiting Research Assistant — *01 – 07/2011 and 02 – 06/2012*
Centre for Transport Studies, Imperial College London — *London, UK*

Research Assistant/Associate — *01/2007 – 04/2013*
Industrial Automation Lab / DELAB, University of Genova — *Genova, Italy*

Education

Doctoral degree in System monitoring and environmental risk management — *01/2010 – 04/2013*
University of Genova — *Genova, Italy*

Master's degree in Computer Engineering - Production and Services Management — *09/2006 – 07/2009*
University of Genova — *Genova, Italy*

Bachelor's degree in Computer Engineering — *09/2003 – 09/2006*
University of Genova — *Genova, Italy*

Participation in Research Projects

At KU Leuven

COCOA - COalition-proof traffic COntrol for Autonomous Vehicles *10/2026 - current*
Funded by the Central Europe Leuven Strategic Alliance - CELSA
Principal Investigator

DREEMS - Optimizing Demand-REsponsive Electric Mobility Systems under uncertainty *01/2026 - current*
Funded by the Research Foundation Flanders - FWO
Co-Principal Investigator

Endowed Chair on Advancing Safe and Efficient Automated Mobility *12-2025 - current*
Funded by Ethias as a 600 years of KU Leuven Chair
Chairholder - Principal Investigator

At Aalto University

ALCOSTO - Adaptive and Learning COntrol strategies for Sustainable future Traffic Operations *09/2022 - current*
Funded by Academy of Finland
Principal Investigator

ACUMEN - Ai-aided deCision tool for seamless mUltiModal nEtwork and traffic managemeNt *06/2023 - 05/2026*
Funded by the European Commission, through the Horizon Europe programme
Coordinator

SMARTEDGE - Semantic Low-code Programming Tools for Edge Intelligence *01/2023 - 03/2026*
Funded by the European Commission, through the Horizon Europe programme
Project Partner Lead

AIForLEssAuto - Artificial Intelligence for Urban Low-Emission Autonomous Traffic *01/2022 - 12/2024*
Funded by Academy of Finland
Subproject Principal Investigator

Finest Twins - Smart Cities Centre of Excellence *12/2019 – 11/2023*
Funded by the European Commission, through the Horizon 2020 programme
Smart mobility stream leader

ULTRA - Ubiquitous Localization, communication, and sensing infrastrucTuRe for Autonomous systems *01/2020 – 12/2022*
Funded by Academy of Finland
Subproject Principal Investigator

Profi4 - Competitive funding to strengthen Universities' research profile *09/2018 – 08/2022*
Funded by Academy of Finland
Smart mobility theme Co-Principal Investigator

CMON - Connected Mobility Now *06/2017 – 06/2022*
Funded by Henry Ford Foundation Finland
Co-Principal Investigator

WISE-ACT - Wider Impacts and Scenario Evaluation of Autonomous and Connected Transport *10/2019 – 03/2022*
Funded by the European Commission, through a COST action
Working Group leader

FDMM - Future Digital Mobility Management *01/2021 – 12/2021*
Funded by EIT Urban Mobility
Project partner Principal Investigator

TRA2020 - The Transport Research Arena 2020 in Helsinki *04/2018 – 10/2020*
Funded by the European Commission, through the Horizon 2020 programme
Work-package leader

MULSIMCO - Multilevel Traffic Simulation with a Cognitive Basis *05/2017 – 08/2018*
Funded by Academy of Finland
Faculty Investigator

SIMSIG - Traffic simulation for intelligent signal control *06/2017 – 05/2018*
Funded by RSM Ltd
Project leader

At Technical University of Crete

TRAMAN21 - Traffic Management for the 21st Century *05/2013 – 02/2018*
Funded by the European Commission, through an ERC Advanced Grant
Postdoctoral Research Associate

At University of Genova

TIP - Transport Integrated Platform *01/2007 – 04/2013*
Funded by Eni R&M - Secondary Logistics
Research assistant / Research associate / Software designer and developer

SECTRAM - Securité du transport de marchandises - Sicurezza nel trasporto merci (Safety for the transport of goods) *04/2009 – 03/2012*
Funded by the European Commission under ALCOTRA (2007-2013)
Research assistant

DaGoT - European Reference Framework on Dangerous Goods Transport *01/2007 – 02/2008*
Funded by the European Commission under the Sixth Framework Programme
Research assistant

Teaching and Educational leadership activities

At KU Leuven

Co-teacher for the Intelligent Transportation Systems course *09/2025 - current*
Part of the Master of Mobility and Supply Chain Engineering

At Aalto University

Director for the Master's programme in Spatial Planning and Transportation Engineering (SPT) course *10/2021 – 09/2024*
Part of the School of Engineering

Deputy Director for the Master's programme in Spatial Planning and Transportation Engineering (SPT) course *08/2019 – 09/2021*
Part of the School of Engineering

Responsible teacher and main instructor for the Transport Modelling course *09/2018 – 02/2024*
Part of the SPT Master's programme

Responsible teacher and main instructor for the Traffic Flow Theory course *10/2016 – 04/2024*
Part of the SPT Master's programme

Deputy responsible teacher and co-instructor for the Traffic Management course *10/2016 – 06/2024*
Part of the SPT Master's programme

Participation in Educational Projects

ENACT - Education network for automated and connected transport for sustainable cities *01/2021 – 07/2023*
Funded by Finnish Ministry of Education and Culture
Project Partner Principal Investigator

Boost I&E 2.0 - Boosting Innovation and Entrepreneurship Challenge Project 2021 *01/2021 – 12/2021*
Funded by EIT Urban Mobility KIC
Activity Lead

Boost I&E - Boosting Innovation and Entrepreneurship Challenge Project *01/2020 – 12/2020*
Funded by EIT Urban Mobility KIC
Activity Partner Principal Investigator

Editorial and organisational activities

Committee memberships

- Member of the Standing Committee on Vehicle-Highway Automation (ACP30) for the Transportation Research Board (TRB) (2022 - 2025)
- Member of the Technical Committee for the IFAC 7.4: Transportation systems (2017 - current)
- Member of the Technical Committee for the IEEE Emerging Transportation Technology Testing (ET3) (2023 - current)
- Representative of the University members in the Governing Board for the European Partnership on Connected, Cooperative & Automated Mobility - CCAM Association (2024 - current)
- Partner Representative of the General Assembly for the European Partnership on Connected, Cooperative & Automated Mobility - CCAM Association (2021 - 2024)
- Vice-chair of the Programme Committee for the Transport Research Arena (TRA2020)
- Member of the Scientific Committee for the Symposia on Management of Future motorway and urban Traffic Systems (MFTS 2016 - 2026)
- Member of the Technical Program Committee for the IFAC Symposia on Control in Transportation Systems (CTS 2018 - 2024)
- Member of the Technical Program Committee for the IEEE International Conference on Intelligent Transportation Systems (ITSC 2017, 2020 - 2023, 2026)
- Member of the Scientific Committee for the European Association for Research in Transportation (hEART 2015 - 2026)
- Member of the Permanent Representative for Aalto University for the European Association for Research in Transportation (hEART 2015 - 2024)
- Member of the Scientific Committee for the IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2021 - 2025)
- Member of the Scientific Committee for the EURO Working Group on Transportation Meetings (EWGT 2016, 2020)
- Member of the International Program Committee for the IEEE Intelligent Vehicles Symposia (IV 2018 - 2022)
- Member of the International Program Committee for the International Conferences on Vehicle Technology and Intelligent Transport Systems (VEHITS 2015 - 2019)
- Member of the Technical Program Committee for the International Conferences on Advances in Vehicular Systems, Technologies and Applications (VEHICULAR 2017 - 2019)
- Member of the Technical Program Committee for the IEEE International Conference on System of Systems (SoSE 2012)

Editor for international journals and conferences

- Editorial Advisory Board Member for Transportation Research Part C: Emerging Technologies
- Associate Editor for IEEE Transactions on Intelligent Transportation Systems
- Special Issues Editor (Deputy Editor-in-Chief) for IET Intelligent Transport Systems
- Associate Editor for IET Intelligent Transport Systems
- Handling Editor for Transportation Research Record: Journal of the Transportation Research Board (TRR)
- Assistant Editor for Journal of Intelligent Transportation Systems (JITS)
- Editor for Proceedings of the Institution of Civil Engineers - Transport
- Review Process Coordinator for TRB Committee on Vehicle-Highway Automation (ACP30)
- Review Editor for Frontiers in Future Transportation
- Principal organiser and Guest Editor of the Special Issue on “Management of Future Motorway and Urban Traffic Systems” within Transportation Research Part C: Emerging Technologies

- Guest Editor of the Special Issue on “Modelling, Learning, and Control of Conventional, Cooperative and Automated Motorway and Urban Traffic Systems” within Transportation Research Part C: Emerging Technologies
- Guest Editor of the Special issue on “Sustainable Development of Intelligent Transportation System and Equipment” within Sustainable Horizons
- Guest Editor of the Special Issue on “Management of Future Motorway and Urban Traffic Systems” within IEEE Transactions on Intelligent Transportation Systems
- Guest Editor of the Special Issue on “Traffic Modelling and Control for Smart, Safe and Green Mobility” within IEEE Open Journal of Intelligent Transportation Systems
- Guest Editor of the Special Issue on “Partially Connected and Automated Traffic Operations in Road Transportation” within Journal of Advanced Transportation
- Guest Editor of the Special issue on “Rethinking transport – towards clean and inclusive mobility” within European Transport Research Review
- Guest Editor of the Special Issue on “Communication, Positioning, and Sensing Solutions for Autonomous Vehicles” within Sensors
- Associate Editor for the IEEE International Conferences on Intelligent Transportation Systems (ITSC)
- Associate Editor for the IFAC Symposia on Control in Transportation Systems (CTS)
- Associate Editor for the IEEE Intelligent Vehicles Symposia (IV)
- Associate Editor for the Forums on Integrated and Sustainable Transportation Systems (Forum ISTS)

Event organisation

- Principal organiser of the 12th Symposium of the European Association for Research in Transportation (hEART 2024)
- Principal organiser of the 1st Symposium on Management of Future motorway and urban Traffic Systems (MFTS 2016)
- Technical programme co-chair of the 8th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2023)
- Member of the Organising Committee for the Transport Research Arena 2020 (TRA2020)
- Co-organiser of the invited session on “Novel methods for modeling and control of mobility and traffic systems” at the 23rd European Control Conference (ECC 2025)
- Organiser of the Breakout session on “Interactive Traffic Management for Highly Automated Vehicles: A Multidisciplinary Challenge” at the 2023 TRB Annual Automated Road Transportation Symposium (ARTS23)
- Co-organiser of the special session titled “Towards sustainable traffic via trajectory optimisation and control strategies for Connected and Automated vehicles” at the 25th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2022)
- Organiser of the Breakout session on “Interactive Traffic Management for Highly Automated Vehicles” at the 2022 TRB Annual Automated Road Transportation Symposium (ARTS22)
- Organiser and moderator of the invited session on “Intelligent Transport Systems” (Nordic Perspective Summit 2020)
- Organiser and Editor of the special Track on “Freeway Traffic Control” at the 15th IFAC Symposium on Control in Transportation Systems (CTS 2018)
- Co-organiser of the open invited track on “Traffic modeling, estimation and control” at the 20th World Congress of the International Federation of Automatic Control (IFAC 2017)
- Co-organiser of the special session on “Recent advances in motorway traffic control” at the 19th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2016)
- Co-organiser of the invited session on “Management and control of motorway traffic systems” at the 14th IFAC Symposium on Control in Transportation Systems (CTS 2016)
- Co-organiser of the special session on “Advances in motorway traffic control” at the 18th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2015)
- Co-organiser of the special session on “Design, modelling, and control of autonomous and cooperative vehicle systems” at the 18th Euro Working Group on Transportation Meeting (EWGT 2015)
- Co-organiser of the invited session on “Optimal control of motorways” at the 20th Conference of the International Federation of Operational Research Societies (IFORS 2014)

Reviewer for international journals

- Transportation Research Part C: Emerging Technologies

- IEEE Transactions on Intelligent Transportation Systems
- Transportation Research Part B: Methodological
- Transportation Research Part A: Policy and Practice
- Transportation Science
- Transportation Research Record
- IET Intelligent Transport Systems
- Automatica
- IEEE Transactions on Control Systems Technology
- Control Engineering Practice
- Annual Reviews in Control
- IEEE Intelligent Transportation Systems Magazine
- IEEE Control Systems Magazine
- International Journal of Robust and Nonlinear Control
- IEEE Systems Journal
- IEEE Control Systems Letters
- Accident Analysis and Prevention
- Journal of Advanced Transportation
- Transportmetrica A: Transport Science
- Transportmetrica B: Transport Dynamics
- Transport Policy
- Transportation Letters: The International Journal of Transportation Research
- Journal of Traffic and Transportation Engineering
- EURO Journal on Transportation and Logistics
- Mathematical and Computer Modelling of Dynamical Systems
- Journal of Intelligent Transportation Systems: Technology, Planning, and Operations
- PROMET – Traffic & Transportation
- European Journal of Transport and Infrastructure Research
- International Journal of Sustainable Transportation
- Physica A
- Advances in Mechanical Engineering

Reviewer for international conferences

- IEEE International Conferences on Intelligent Transportation Systems (ITSC)
- IEEE Intelligent Vehicles Symposia (IV)
- IEEE Conferences on Decision and Control (CDC)
- European Control Conferences (ECC)
- IFAC World Congresses (WS)
- IFAC Symposia on Control in Transportation Systems (CTS)
- Transportation Research Board (TRB) Annual Meetings
- Symposia of the European Association for Research in Transportation (hEART)
- EURO Working Group on Transportation Meetings (EWGT)
- International Symposia on Transportation and Traffic Theory (ISTTT)
- international Conferences on Traffic and Granular Flow (TGF)
- International Conferences on Vehicle Technology and Intelligent Transport Systems (VEHITS)
- European Conference on Information Systems (ECIS)
- IEEE International Conferences on System of Systems (SoSE)
- International Conferences on System Theory, Control and Computing (ICSTCC)

Supervision activities

At KU Leuven

Ms. Ayşenaz İpek Gülcü, PhD student *10/2026 – current*
KU Leuven, Belgium

Mr. Yuangxiang Yang, PhD student *10/2025 – current*
KU Leuven, Belgium - also at Aalto University, Finland

Mr. Hongwei Li, PhD student *09/2025 – current*
KU Leuven, Belgium - also at Tel Aviv University, Israel

Mr. Matteo Paiano, PhD student *03/2025 – current*
KU Leuven, Belgium

Mr. Hao Lyu, PhD student *10/2025 – current*
Southeast University, China

Mr. Franco Filippi, PhD student *01/2026 – 06/2026*
University of Salerno, Italy

Ms. Yanqing Yang, PhD student *05/2025 – 04/2026*
Tongji University, China

Mr. Seshadri Naik Moode, PhD student *09/2025 – 01/2026*
Universitat Politècnica de Catalunya, Spain

At Aalto University

Ms. Liva Westerback, PhD student *11/2023 – current*
Aalto University, Finland

Ms. Heqi Wang, PhD student *03/2023 – current*
Aalto University, Finland

Mr. Muhammad Haris, PhD student *09/2022 – current*
Aalto University, Finland

Ms. Elina Aittoniemi, Part-time PhD student *11/2018 – current*
Aalto University, Finland - also at VTT Technical Research Centre of Finland

Mr. Francesco Vitale, PhD student & Postdoc *01/2020 – 07/2026*
Aalto University, Finland

Mr. Yuanxiang Yang, PhD student *10/2022 – 10/2025*
Aalto University, Finland - moved to KU Leuven

Mr. Serio Agriesti, PhD student *03/2020 – 10/2024*
Aalto University, Finland

Mr. Ze Zhou, PhD student *04/2019 – 10/2024*
Aalto University, Finland

Mr. Roozbeh Mohammadi, PhD student & Postdoc *02/2018 – 08/2023*
Aalto University, Finland

Mr. Farzam Tajdari, Part-time PhD student *10/2017 – 01/2023*
Aalto University, Finland

Dr. Iisakki Kosonen, Senior researcher *04/2019 – 10/2024*
Aalto University, Finland

Dr. Murat Bayrak, Postdoctoral researcher *08/2023 – 12/2025*
Aalto University, Finland

Dr. Shaghayegh Vosough, Postdoctoral researcher *03/2021 – 07/2025*
Aalto University, Finland

Dr. Charalampos Sipetas, Postdoctoral researcher *01/2021 – 01/2025*
Aalto University, Finland

Dr. Ramin Niroumand, Postdoctoral researcher *08/2023 – 08/2024*
Aalto University, Finland

Dr. Weiming Zhao, Postdoctoral researcher *11/2019 – 09/2022*
Aalto University, Finland

Dr. Cafer Avci, Postdoctoral researcher *06/2019 – 06/2022*
Aalto University, Finland

Dr. Yu Han, Postdoctoral researcher *09/2017 – 10/2017*
Aalto University, Finland

Mr. Jingfeng Ma, PhD student *12/2022 – 12/2023*
Southeast University, China

Ms. Haizhen Li, PhD student *03/2021 – 03/2022*
Chang'an University, China

Mr. Mohammad Karimi, PhD student *08/2017 – 06/2020*
Concordia University, Canada

Ms. Oxana Ivanova, MSc student *11/2024 – 12/2025*
Aalto University, Finland

Ms. Julia Latvala, MSc student *11/2024 – 05/2025*
Aalto University, Finland

Mr. Chris Yung, MSc student *06/2024 – 11/2024*
Aalto University, Finland

Mr. Germán R. Pardo González, MSc student *06/2024 – 08/2024*
Aalto University, Finland - also at London School of Economics, UK

Ms. Olena Vol, Summer intern *06/2022 – 08/2022*
Aalto University, Finland - also at Université Paris 1 Panthéon-Sorbonne, France

Mr. Petr Anashin, Summer intern *06/2022 – 08/2022*
Aalto University, Finland - also at University of Tartu, Estonia

Mr. Héctor Del Pino Verona, MSc student *09/2019 – 01/2020*
Aalto University, Finland - also at Universitat Politècnica de València, Spain

Mr. Petr Hajduk, MSc student *01/2018 – 12/2018*
Aalto University, Finland - also at VTT Technical Research Centre of Finland

At Technical University of Crete

Ms. Maria Kontorinaki, PhD student *09/2017 – 09/2017*
Technical University of Crete, Greece - with Prof. Markos Papageorgiou

Ms. Georgia R. Iordanidou, PhD student *06/2013 – 03/2017*
Technical University of Crete, Greece - with Prof. Ioannis Papamichail

Ms. Sofia Papadopoulou, MsC student *01/2016 – 03/2017*
Technical University of Crete, Greece - with Prof. Markos Papageorgiou

Ms. Georgia Perraki, MsC student *03/2015 – 11/2016*
Technical University of Crete, Greece - with Prof. Ioannis Papamichail

Mr. Markos Fountoulakis, MsC student *03/2015 – 07/2016*
Technical University of Crete, Greece - with Prof. Markos Papageorgiou

Ms. Cecilia Pasquale, PhD student *05/2014 – 07/2014*
University of Genova, Italy - with Prof. Simona Sacone

At University of Genova

Mr. Haitam Laarabi, PhD student *07/2012 – 04/2013*
University of Genova, Italy - with Prof. Roberto Sacile

Invited presentations

Network-wide motorway traffic control in presence of Vehicle Automation and Communication Systems *26/10/2015*
University of California Los Angeles, USA
IPAM Workshop on Traffic Control, Part of the long program "New directions in mathematical approaches for traffic flow management"

Traffic management in the era of Vehicle Automation and Communications Systems and Smart Infrastructure *09/12/2015*
University of Glasgow, UK
Invited lecture

Multilane traffic control in the presence of automated and connected vehicles *01/06/2017*
University of Luxembourg, Luxembourg
Workshop on Smart Mobility

Integrated control of motorway bottlenecks via flow metering and lane assignment *12/09/2018*
Technical University of Crete, Greece
Invited seminar

Traffic management for CAVs: Influence on traffic characteristics and control strategies *28/05/2019*
Beijing University of Technology, China
Invited seminar

User-based signal timing in a connected vehicle environment *31/05/2019*
Zhejiang University, China
Invited seminar

Feedback-based integrated lane-based control with Connected and Automated Vehicles *06/11/2019*
University of Canterbury, New Zealand
Invited seminar

Data-driven traffic flow estimation at intersections in a partially connected environment *20/09/2020*
IEEE Intelligent Transportation System Conference 2020, Greece
Workshop on Traffic Management for Future Mobility: CAVs in a Mixed Traffic Environment

Traffic Management in the era of Connectivity and Automation *08/10/2020*
TU Dresden, Germany
Invited lecture

Online set-point estimation for motorway traffic control with connected and automated vehicles *26/10/2022*
University of Cyprus, Cyprus
Invited seminar

Integration of large-scale activity-based and traffic assignment models: Calibration and equilibrium *18/10/2023*
Queensland University of Technology, Australia
Invited seminar

Connected and Automated Driving: Obstacle avoidance and cooperative decentralised routing *23/10/2023*
University of Queensland, Australia
Invited seminar

Connected and Automated Driving: Decentralised signal-free intersection management and cooperative routing *03/11/2023*
University of Sydney, Australia
Invited seminar

Dynamic ridesharing: Considering traffic congestion effects and fair pricing in matching and routing *13/11/2023*
University of New South Wales, Australia
Invited seminar

Matching and routing in Dynamic Ridesharing: Addressing traffic congestion and fair pricing *21/03/2024*
TU Delft, the Netherlands
Invited webinar

Controlling traffic when flow dynamics evolve *16/06/2026*
Traffic and Granular Flow 2026, UK
Keynote speaker

Invited external examiner

PhD degree examination *12/2025*
University of Alcalá, Spain

PhD degree examination *09/2025*
Technical University of Crete, Greece

Faculty evaluator *02/2025*
Université Gustave Eiffel, France

PhD degree examination *09/2025*
Technical University of Crete, Greece

PhD degree examination *11/2024*
University of Tartu, Estonia

PhD degree examination *05/2024*
Technion – Israel Institute of Technology, Israel

PhD degree examination *05/2024*
Trinity College Dublin, Ireland

PhD degree examination *12/2022*
Technical University of Crete, Greece

PhD entry examination *12/2020*
Swiss Federal Institute of Technology in Zurich (ETH), Switzerland

PhD degree examination *07/2019*
Technical University of Crete, Greece

PhD degree examination *09/2017*
Technical University of Crete, Greece

PhD degree examination *03/2017*
Technical University of Crete, Greece

2 MSc degree theses *02/2017*
École polytechnique fédérale de Lausanne (EPFL), Switzerland

Research evaluation

- Evaluator of research proposals for California METRANS Transportation Center
- Evaluator of research proposals for the Netherlands Organisation for Scientific Research (NWO), domain Applied and Engineering Sciences
- Evaluator of research proposals for the Luxembourg national research fund (FNR)
- Evaluator of research proposals for the Research Grants Council (RGC) of Hong Kong
- Evaluator of research proposals for the Swiss National Science Foundation (SNSF)
- Evaluator of grant proposals for the National Science Center (NCN) of Poland
- External Expert in the evaluation of COST Action proposals, for the European Commission

Recognitions and awards

- The paper by Roncoli, C., Papageorgiou, M., Papamichail, M., titled "Traffic flow optimisation in presence of vehicle automation and communication systems - Part II: Optimal control for multi-lane motorways", was listed among the "Most Cited Transportation Research Part C: Emerging Technologies Articles" during 2015-2018.
- The paper by Spiliopoulou, A., Perraki, G., Papageorgiou, M., Roncoli, C., titled "Exploitation of ACC systems towards improved traffic flow efficiency on motorways", which was presented by A. Spiliopoulou at the 5th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2017), Naples, Italy, June 26-28, 2017, received the "Best Paper Presented by a Woman award".
- The paper by Bekiaris-Liberis, N., Roncoli, C., Papageorgiou, M., titled "Highway Traffic State Estimation With Mixed Connected and Conventional Vehicles", received the 2019 "George N. Saridis Best Transactions Paper Award" for Outstanding Research.
- The paper by Zhou, Z., Roncoli, C., titled "A fairness-aware joint pricing and matching framework for dynamic ridesharing", which was presented by Z. Zhou at the 8th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2023), Nice, France, June 14-16, 2023, received the "Best Paper Award".
- The paper by Vitale, F., Roncoli, C., titled "Real-time congestion-aware distributed cooperative rerouting of connected and automated vehicles in urban networks", which was presented by F. Vitale at the TRB 103rd Annual Meeting, Washington, DC, USA, January 7-11, 2024, received the "Best Young First Author Award" for the Operations Section.

Personal information

- Nationality: Italian
- Date of birth: 30/10/1984
- Languages: Italian (mother tongue), English (fluently), Dutch (intermediate), French (basic), Greek (basic)

Publication list

Articles in international journals

1. Roncoli, C., Bersani, C., and Sacile, R. A risk-based system of systems approach to control the transport flows of dangerous goods by road. *IEEE Systems Journal*, 7(4):561–570, 2013. DOI: 10.1109/JSYST.2012.2212652.
2. Iordanidou, G.-R., Roncoli, C., Papamichail, I., and Papageorgiou, M. Feedback-based mainstream traffic flow control for multiple bottlenecks on motorways. *IEEE Transactions on Intelligent Transportation Systems*, 16 (2):1–12, 2014. DOI: 10.1109/TITS.2014.2331985.
3. Pasquale, C., Papamichail, I., Roncoli, C., Sacone, S., Siri, S., and Papageorgiou, M. Two-class freeway traffic regulation to reduce congestion and emissions via nonlinear optimal control. *Transportation Research Part C: Emerging Technologies*, 55(1):85–99, 2015. DOI: 10.1016/j.trc.2015.01.013.
4. Roncoli, C., Papageorgiou, M., and Papamichail, I. Traffic flow optimisation in presence of vehicle automation and communication systems – Part I: A first-order multi-lane model for motorway traffic. *Transportation Research Part C: Emerging Technologies*, 57:241–259, 2015. DOI: 10.1016/j.trc.2015.06.014.
5. Roncoli, C., Papageorgiou, M., and Papamichail, I. Traffic flow optimisation in presence of vehicle automation and communication systems – Part II: Optimal control for multi-lane motorways. *Transportation Research Part C: Emerging Technologies*, 57:260–275, 2015. DOI: 10.1016/j.trc.2015.05.011.
6. Roncoli, C., Papamichail, I., and Papageorgiou, M. Hierarchical model predictive control for multi-lane motorways in presence of Vehicle Automation and Communication Systems. *Transportation Research Part C: Emerging Technologies*, 62:117–132, 2016. DOI: 10.1016/j.trc.2015.11.008.
7. Bekiaris-Liberis, N., Roncoli, C., and Papageorgiou, M. Highway traffic state estimation with mixed connected and conventional vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 17(12):3484–3497, 2016. DOI: 10.1109/TITS.2016.2552639.
8. Roncoli, C., Bekiaris-Liberis, N., and Papageorgiou, M. Use of speed measurements for highway traffic state estimation: Case studies on NGSIM data and highway A20, Netherlands. *Transportation Research Record: Journal of the Transportation Research Board*, 2559(1):90–100, 2016. DOI: 10.3141/2559-11.
9. Han, Y., Hegyi, A., Yuan, Y., Hoogendoorn, S., Papageorgiou, M., and Roncoli, C. Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits. *Transportation Research Part C: Emerging Technologies*, 77:405–420, 2017. DOI: 10.1016/j.trc.2017.02.009.
10. Fountoulakis, M., Bekiaris-Liberis, N., Roncoli, C., Papamichail, I., and Papageorgiou, M. Highway traffic state estimation with mixed connected and conventional vehicles: Microscopic simulation-based testing. *Transportation Research Part C: Emerging Technologies*, 78:13–33, 2017. DOI: 10.1016/j.trc.2017.02.015.
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